PACK INSERT FOR MALAYSIA

Kytril® Tablet

ATNAHS

Granisetron

Antiemetics, Serotonin (5HT3) antagonists

1. Pharmaceutical Form

Kytril is supplied as film-coated tablets and ampoules

2. Qualitative and Quantitative Composition

Active ingredient: granisetron HCL
White triangular film-coated tablets contain 1 mg of granisetron (free base equivalent).

3. Clinical Particulars
3.1 Therapeutic Indications
KYTRIL is indicated for the prevention and treatment (control) of acute and delayed nausea and vomiting associated with chemo-therapy and radiotherapy
3.2 Dosage and Method of Administration
3.2.1 Standard Dosage
Chemotherapy Induced Nausea and Vomiting (CINV)
Adults
Prevention: 1 mg twice a day or 2 mg once a day for up to one week following chemotherapy. The first dose of Kytril should be administered within 1 hour before the start of therapy.
Treatment: There is insufficient information to recommend the oral administration of KYTRIL in the treatment of CINV in adult patients.
Pediatrics
There is insufficient information to recommend the oral admin-istration of KYTRIL in the prevention and treatment of CINV in pediatric patients.
Radiotherapy Induced Nausea and Vomiting (RINV)
Adults
Prevention and treatment: 2 mg once a day for up to one week following radiotherapy. The first dose of KYTRIL should be ad-ministered within 1 hour before the start of therapy.
Pediatrics
There is insufficient information to recommend oral administra-tion of Kytril in the prevention and treatment of RINV in children.
3.2.2 Special Dosage Instruction
Geriatrics: No dosage adjustments required.
Renal impairment: No dosage adjustments required.
Hepatic Impairment: No dosage adjustments required.
3.3 Contraindications
Kytril is contraindicated in patients with known hypersensitivity to granisetron or to any of its excipients.
3.4 Special Warnings and Special Precautions for Use
As Kytril may reduce lower bowel motility, patients with signs of sub-acute intestinal obstruction should be monitored closely fol-lowing administration of Kytril.
As with other 5-HT3 antagonists, cases of ECG modifications including QT prolongation have been reported with KYTRIL. These ECG changes with KYTRIL were minor and generally not of clinical significance, specifically with no evidence of proarrhyth-mia. However, in patients with pre-existing arrhythmias or cardiac conduction disorders, this might lead to clinical consequences. Therefore, caution should be exercised in patients with cardiac co-morbidities, on cardio-toxic chemotherapy and/or with concomi-tant electrolyte abnormalities.
Cross-sensitivity between 5-HT3 antagonists has been reported.
It is recommended that Kytril film-coated tablets are not taken by patients with rare hereditary problems of galactose intolerance, lactase deficiency or glucose-galactose malabsorption.
As with other 5-HT3 antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neu-romus-cular abnormalities) have been reported following the con-comitant use of Kytril and other serotonergic drugs. If concomi-tant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of this patient is advised.
3.5 Interactions with other Medicinal Products and Other Forms of Interaction
Kytril did not induce or inhibit the cytochrome P450 drug metabo-liz-ing enzyme system in rodent studies or inhibit the activity of any well characterized P450 sub-families studied in in vitro inves-tigations.
In humans, hepatic enzyme induction with phenobarbital resulted in an increase in total plasma clearance of intravenous Kytril of approximate-ly one-quarter. In in vitro human microsomal studies, ketoconazole inhibited ring oxidation of Kytril. However, given the absence of pK/pD relationship with granisetron, these changes are believed to have no clinical consequences.
Kytril has been safely administered in humans with benzodiaze-pines, neuroleptics and anti-ulcer medications, commonly pre-scribed with antiemetic treatments. Additionally, Kytril has shown no apparent drug interaction with emetogenic cancer chemothera-pies.
No specific interaction studies have been conducted in anesthe-tized patients, but Kytril has been safely administered with com-monly used anesthetic and analgesic agents. In addition, the activi-ty of the cytochrome P450 subfamily 3A4 (involved in the metab-olism of some of the main narcotic analgesic agents) is not modi-fied by Kytril.
As for other 5-HT3 antagonists, cases of ECG modifications in-cluding QT prolongation have been reported with KYTRIL. These ECG changes with KYTRIL were minor and generally not of clini-cal significance, specifically with no evidence of proarrhythmia. However, in patients concurrently treated with drugs known to prolong QT interval and/or are arrhythmogenic, this may lead to clinical consequences.
As with other 5-HT3 antagonists, cases of serotonin syndrome have been reported following the concomitant use of Kytril and other serotonergic drugs. If concomitant treatment with grani-setron and other serotonergic drugs is clinically warranted, appro-priate observa-tion of this patient is advised (see 3.4 Special Warn-ings and Special Precautions for Use).
3.6 Pregnancy and Lactation
There are no studies in pregnant women and it is not known whether granisetron is excreted in human milk. Use of Kytril dur-ing pregnancy or lactation should be limited to situations where the potential benefit to the mother justifies the potential risk to the fetus or nursing infant (see 4.2.6 Preclinical Safety).
3.7 Effects on Ability to Drive and Use Machines
In healthy subjects, no clinically relevant effects on resting EEG or on the performance of psychometric tests were observed after i.v. Kytril at any dose tested (up to 200 µg/kg). There are no data on the effect of Kytril on the ability to drive or use machinery.
3.8 Undesirable Effects
3.8.1 Clinical Trials
Summary of the safety profile
The most frequently reported adverse reactions for Kytril are headache and constipation which may be transient. ECG changes including QT prolongation have been reported with Kytril (see sections 3.4 and 3.5).
The following table of listed adverse reactions is derived from clinical trials and post-marketing data associated with Kytril.
Frequency categories are as follows:
Very common: ≥1/10;
Common ≥1/100 to <1/10;
Uncommon ≥1/1,000 to <1/100
Rare ≥1/10,000 to <1/1,000)
Very rare (<1/10,000)
Table 1 Tabulated List of Adverse Reactions

Immune system disorders	
Uncommon	Hypersensitivity reactions e.g. anaphylaxis, urticaria
Nervous system disorders	
Very common	Headache
Uncommon	Serotonin Syndrome
Cardiac disorders	
Uncommon	QT prolongation
Gastrointestinal disorders	
Very common	Constipation
Hepatobiliary disorders	

Common	Elevated hepatic transaminases*
Skin and subcutaneous tissue disorders	
Uncommon	Rash

*Occurred at a similar frequency in patients receiving comparator therapy

Kytril has been well tolerated in human studies. In common with other drugs of this class, headache and constipation have been reported. Cases of hypersensitivity reactions, including rashes and anaphylaxis have been reported. Elevations in hepatic transami-nases have been observed and at similar frequency in patients receiving comparator therapy.
As for other 5-HT3 antagonists, cases of ECG modifications in-cluding QT prolongation have been reported with KYTRIL. These ECG changes with KYTRIL were minor and generally not of clini-cal significance, specifically with no evidence of proarrhythmia. (See 3.4 Warnings and Precautions, General and 3.5 Interactions with other Medicinal Products and other Forms of Interaction)
As with other 5-HT3 antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neu-romus-cular abnormalities) have been reported following the concomitant use of Kytril and other serotonergic drugs (see 3.4 Special Warnings and Special Precautions for Use and 3.5 Interac-tions with other Medicinal Products and other Forms of Interac-tion) .

3.8.2 Post Marketing

The post-marketing safety experience in over 4 million patients is consistent with the clinical trial safety information.
For ECG modifications, see 3.8 Undesirable Effects, Clinical Tri-als.
For serotonin syndrome, see 3.8 Undesirable Effects, Clinical Trials.

3.9 Overdose

There is no specific antidote for Kytril. In the case of overdosage with Kytril, symptomatic treatment should be given. Overdosage of up to 38.5 mg of granisetron hydrochloride as a single injection has been reported without symptoms or only the occurrence of a slight headache.

4. Pharmacological Properties & Effects

4.1 Pharmacodynamic Properties

4.1.1 Mechanism of Action

Serotonin receptors of the 5-HT3 type are located peripherally in vagal nerve terminals and centrally in the chemoreceptor trigger zone of the area postrema. During chemotherapy-induced vomit-ing, mucosal enterochromaffin cells release serotonin, which stim-ulates 5 HT3 receptors. This invokes vagal afferent discharge, inducing vomiting.
Kytril is a potent anti-emetic and highly selective antagonist of 5-hydroxytryptamine (5 HT3) receptors. Radioligand binding studies have demonstrated that Kytril has negligible affinity for other re-ceptor types including 5-HT and dopamine D2 binding sites.

4.1.2 Efficacy / Clinical Studies

Chemotherapy-induced nausea and vomiting(CINV)

KYTRIL administered intravenously has been shown to be effec-tive in the prevention and treatment of nausea and vomiting asso-ciated with cancer chemotherapy in adults. Kytril administered orally has been shown to be effective in the prevention of nausea and vomiting associated with cancer chemotherapy. Kytril admin-istered intravenously has been shown to be effective in children aged 2 years and above for the prevention and treatment of acute CINV. There is insufficient information to recommend the oral administration of KYTRIL in the prevention and treatment of CINV in pediatric patients.

Radiation-induced nausea and vomiting(RINV)

Kytril administered orally has been shown to be effective in the prevention and treatment of nausea and vomiting associated with total body or fractionated abdominal irradiation in adults. Efficacy in children has not been established in controlled clinical trials.

Postoperative nausea and vomiting(PONV)

Kytril administered intravenously has been shown to be effective for prevention and treatment of post-operative nausea and vomit-ing in adults.

4.2 Pharmacokinetic Properties

4.2.1 Absorption

Absorption of Kytril is rapid and complete, though oral bioavaila-bility is reduced to about 60% as a result of first pass metabolism. Oral bioavailability is generally not influenced by food.

4.2.2 Distribution

Kytril is extensively distributed, with a mean volume of distribu-tion of approximately 3l/kg. Plasma protein binding is approxi-mately 65%.

4.2.3 Metabolism

Biotransformation pathways involve N-demethylation and aro-matic ring oxidation followed by conjugation. In-vitro liver micro-somal studies show that granisetron's major route of metabolism is inhibited by ketoconazole, suggestive of metabolism mediated by the cytochrome P-450 3A subfamily.

4.2.4 Elimination

Clearance is predominantly by hepatic metabolism. Urinary excre-tion of unchanged Kytril averages 12% of dose while that of me-tabolites amounts to about 47% of dose. The remainder is excreted in feces as metabolites. Mean plasma half-life in patients by the oral and intrave-nous route is approximately 9 hours, with a wide inter-subject variability.
The pharmacokinetics of oral and intravenous Kytril demonstrate no marked deviations from linear pharmacokinetics at oral doses up to 2.5-fold and intravenous doses up to 4-fold the recommend-ed clinical dose.
The results of a study in healthy male volunteers have demon-strated that systemic delivery of 3 mg granisetron from an intra-muscular injection is slower than from a 5 minute intravenous infusion (as indicated by lower Cmax and later Tmax). In other re-spects, the pharmacokinetics of granisetron are virtually indistin-guishable when administered by these two different routes.

4.2.5 Pharmacokinetics in Special Populations

Renal failure: In patients with severe renal failure, data indicate that pharmacokinetic parameters after a single intravenous dose are generally similar to those in normal subjects.
Hepatic impairment: In patients with hepatic impairment due to neoplastic liver involvement, total plasma clearance of an intrave-nous dose was approximately halved compared to patients without hepatic involvement. Despite these changes, no dosage adjustment is necessary.
Elderly: In elderly subjects after single intravenous doses, phar-macoki-netic parameters were within the range found for non-elderly subjects.
Pediatrics: In children, after single intravenous doses, pharmaco-kinet-ics are similar to those in adults when appropriate parameters (volume of distribution, total plasma clearance) are normalized for body weight.

4.2.6 Preclinical Safety

Preclinical data revealed no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxici-ty, reproductive toxicity and genotoxicity. Carcinogenicity studies revealed no special hazard for humans when used in the recom-mended human dose. However, when administered in higher dos-es and over a prolonged period of time the risk of carcinogenicity cannot be ruled out.
In carcinogenicity studies in rats and mice treated orally for their lifetime (2 years), no adverse findings were observed at dosages 25 times the clinical dose. At higher doses, KYTRIL induced cell proliferation in the rat liver and hepatocellular tumors in rats and mice.
KYTRIL was not mutagenic in mammalian or non-mammalian in vivo or in vitro test systems and there was no evidence of un-scheduled DNA synthesis indicating that KYTRIL is not genotox-ic.
In the rat, KYTRIL had no untoward effect on reproductive per-formance, fertility or on pre- and post natal development.
Teratogenic effects were not observed in rats or rabbits.

5. Pharmaceutical Particulars

5.1 List of Excipients

Tablets

Lactose

5.2 Stability

This medicinal product should not be use after the expiry date (EXP) shown on the outer pack.

5.3. Special Precautions for Storage

Do not store above 30˚C, see also outer pack for storage remarks.

6. Packs

Ampoules, 1mg/1ml, 3mg/3ml	5, 5
Tablets, 1mg	2, 10

Medicine: Keep out of reach of children

MYKytrilTab0714/CDS3.0

ATNAHS

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Film-coated tablets
Made for
Atnahs Pharma UK Ltd,
Basildon, United Kingdom.
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